

ALyl Chem 15 EQ P1 22w to 09s Paper 1 Halogen compounds 65marks

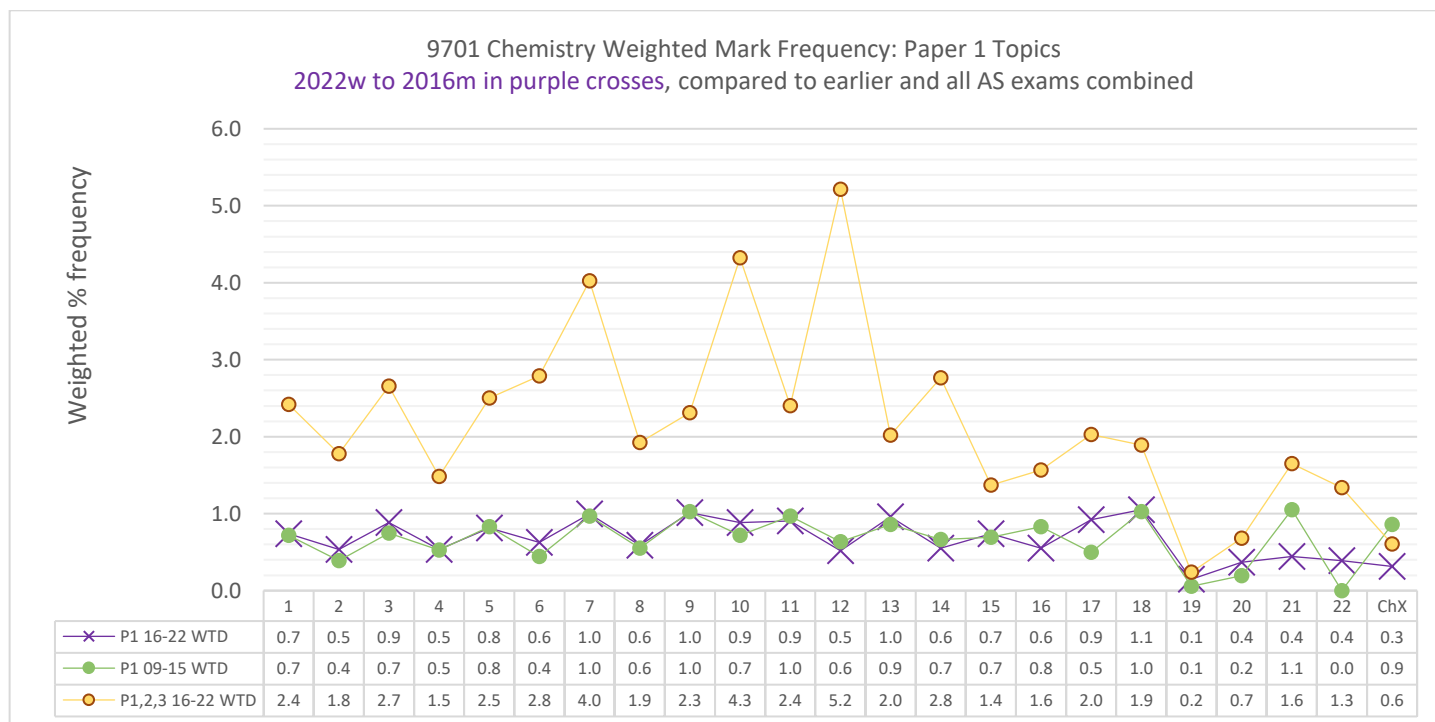
As you start and work through this worksheet you can tick off your progress to show yourself how much you have done, and what you need to do next. The first task is just to read the first question and should take you less than one minutes to complete.

Paper 1 Topic 15

Checklist Tick each task off as you go along

RANK:

P1 Noob	P1 Novice	P1 Bronze	P1 Silver	P1 Gold	P1 ¹ Winner	P1 Hero	P1 Legend
1 Q started	1 Q done	10% of marks	25% of marks	40% of marks	50% of marks	75% of marks	100% of marks
	1	7	16	26	33	49	65
	1	8	20	33	41	61	81



What the most thoughtful students will get out of their extensive studying will be a capacity to do meaningful brain-based work even under stressful conditions, which is a part of the self-mastery skillset that will continue to deliver value for the whole of their lives. Outstanding grades will also happen, but the most important goal from skillful action in study is being better at any important task, even if circumstances do not feel ideal.

As you are moving through your studies you can learn more about yourself by trying out new ways to manage yourself, and analysing how effective those new techniques were. In this reflective process not only will you get better at working positively and productively to deliver ambitious and successful outcomes, but you will be working towards one aspect of life's highest pursuit, summarised and inscribed on the Temple of Apollo at Delphi: "know thyself".

1. To complete these questions, as important as your answer, is checking your answer against the mark scheme.
2. For each page or group of 10 questions, convert your mark score into a percentage. This will allow you to see (and feel) your progress as you get more experience and understanding with each topic.
3. Multiple choice questions, done carefully where you explain and show yourself your thinking using written notes as you move through each question, can be more useful than just Paper 2 for students aiming for a C or B grade. Paper 2 should be the larger focus for students aiming for A and A* grades, however.
4. If you find you get a higher percentage answering short answer questions than multiple choice questions that often means you are NOT using the marking scheme correctly; your correct answer might not be fully complete for all the marks you are awarding. The marks easiest to miss rely on providing the largest amount of detail.

¹ **DO NOT** work on these higher levels of completion in your A2 year unless you have also achieved at least a "Silver" (25%) in the same topic in **Paper 2**, which is **MOST** of your **AS grade**, and Paper 3 which is a smaller part of your year but still important.

15 Halogen compounds

15.1 Halogenoalkanes

Learning outcomes

Candidates should be able to:

- recall the reactions (reagents and conditions) by which halogenoalkanes can be produced:
 - the free-radical substitution of alkanes by Cl_2 or Br_2 in the presence of ultraviolet light, as exemplified by the reactions of ethane
 - electrophilic addition of an alkene with a halogen, X_2 , or hydrogen halide, HX(g) , at room temperature
 - substitution of an alcohol, e.g. by reaction with HX or KBr with H_2SO_4 or H_3PO_4 ; or with PCl_3 and heat; or with PCl_5 ; or with SOCl_2
- classify halogenoalkanes into primary, secondary and tertiary
- describe the following nucleophilic substitution reactions:
 - the reaction with NaOH(aq) and heat to produce an alcohol
 - the reaction with KCN in ethanol and heat to produce a nitrile
 - the reaction with NH_3 in ethanol heated under pressure to produce an amine
 - the reaction with aqueous silver nitrate in ethanol as a method of identifying the halogen present as exemplified by bromoethane

15.1 Halogenoalkanes (continued)

- describe the elimination reaction with NaOH in ethanol and heat to produce an alkene as exemplified by bromoethane
- describe the $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ mechanisms of nucleophilic substitution in halogenoalkanes including the inductive effects of alkyl groups
- recall that primary halogenoalkanes tend to react via the $\text{S}_{\text{N}}2$ mechanism; tertiary halogenoalkanes via the $\text{S}_{\text{N}}1$ mechanism; and secondary halogenoalkanes by a mixture of the two, depending on structure
- describe and explain the different reactivities of halogenoalkanes (with particular reference to the relative strengths of the C-X bonds as exemplified by the reactions of halogenoalkanes with aqueous silver nitrates)

Changes to this syllabus for 2025, 2026 and 2027

The syllabus has been updated. This is version 1, published September 2022.

You must read the whole syllabus before planning your teaching programme. We review our syllabuses regularly to make sure they continue to meet the needs of our schools. In updating this syllabus, we have made it easier for teachers and students to understand, keeping the familiar features that teachers and schools value.

Changes to syllabus content

- Update to reagents in 15.1(c) and 16.2(b).

Any textbooks endorsed to support the syllabus for examination from 2022 are still suitable for use with this syllabus.



15 Halogen compounds**15.1 Halogenoalkanes****Learning outcomes**

Candidates should be able to:

- 1 recall the reactions (reagents and conditions) by which halogenoalkanes can be produced:
 - (a) the free-radical substitution of alkanes by Cl_2 or Br_2 in the presence of ultraviolet light, as exemplified by the reactions of ethane
 - (b) electrophilic addition of an alkene with a halogen, X_2 , or hydrogen halide, HX(g) , at room temperature
 - (c) substitution of an alcohol, e.g. by reaction with HX(g) ; or with KCl and concentrated H_2SO_4 or concentrated H_3PO_4 ; or with PCl_3 and heat; or with PCl_5 ; or with SOCl_2

Q# 935/ AS Chemistry/2022/w/TZ 1/Paper 1/Q# 37//www.SmashingScience.org :o)

37 2-bromopropane reacts with hot ethanolic sodium hydroxide.

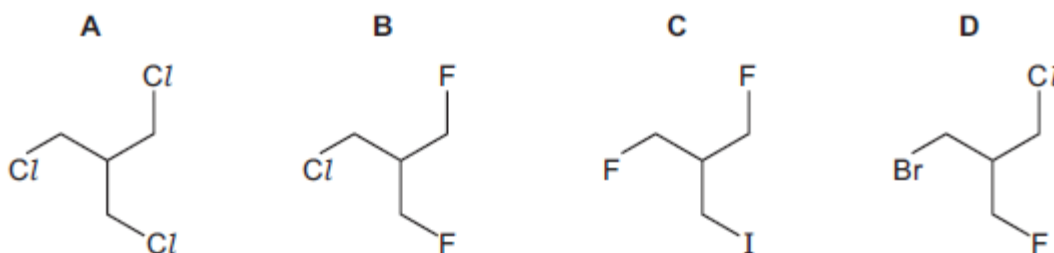
Which substance is the major product of this reaction?

- A propan-1-ol
 B propan-2-ol
 C 2-hydroxypropene
 D propene

Q# 936/ AS Chemistry/2022/s/TZ 1/Paper 1/Q# 32//www.SmashingScience.org :o)

32 The presence of a halogen in an organic compound may be detected by warming the organic compound with aqueous silver nitrate.

Which compound would be the quickest to produce a precipitate?



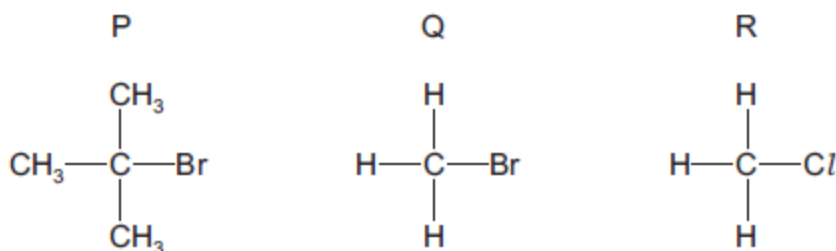
Q# 937/ AS Chemistry/2022/s/TZ 1/Paper 1/Q# 31//www.SmashingScience.org :o)

31 Structural isomerism and stereoisomerism should be taken into account when answering this question.How many isomeric alkenes with formula C_5H_8 are present in the mixture produced when 1,4-dibromopentane is reacted with NaOH in ethanol?

- A 1 B 2 C 3 D 4



30 The diagram shows the structures of three halogenoalkanes.



P, Q and R can all be hydrolysed.

Which row is correct?

	relative speed of hydrolysis		mechanism of hydrolysis	
	Q	R	P	Q
A	fast	slow	S _N 1	S _N 2
B	fast	slow	S _N 2	S _N 1
C	slow	fast	S _N 1	S _N 2
D	slow	fast	S _N 2	S _N 1

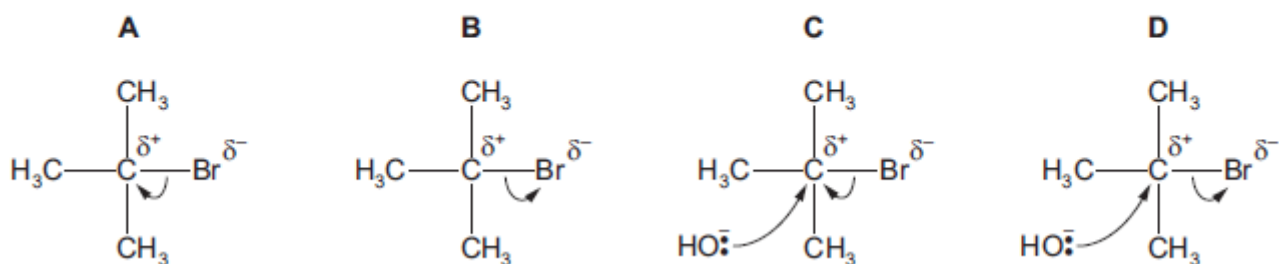
29 A reaction occurs when a sample of 1-chloropropane is heated under reflux with sodium hydroxide dissolved in ethanol.

Which row is correct?

	type of reaction	name of product
A	elimination	propan-1-ol
B	elimination	propene
C	substitution	propan-1-ol
D	substitution	propene

23 When 2-bromo-2-methylpropane reacts with aqueous sodium hydroxide, an alcohol is formed.

Which diagram describes the first step in the reaction mechanism?



22 Structural isomerism **and** stereoisomerism should be considered when answering this question.

A colourless liquid, $C_5H_{11}Cl$, exists as a mixture of two optical isomers.

When heated with sodium hydroxide in ethanol, a mixture of **only two** alkenes is formed.

What could the colourless liquid be?

- A $(CH_3CH_2)_2CHCl$
- B $CH_3CH_2CH_2CHClCH_3$
- C $CH_3CH_2CCl(CH_3)_2$
- D $(CH_3)_2CHCHClCH_3$

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

37 Which statements are correct?

- 1 1,1-difluoroethane is less reactive than 1,1-dichloroethane.
- 2 1,1-difluoroethane is polar.
- 3 The C–F bond is stronger than the C–Cl bond.

20 Bromoethane reacts with cyanide ions, producing propanenitrile.

Which statement about the S_N2 mechanism of this reaction is correct?

- A The lone pair of electrons on C of CN^- attacks the carbon atom of the C–Br bond.
- B The lone pair of electrons on C of CN^- attacks the carbocation formed when the C–Br bond breaks.
- C The lone pair of electrons on N of CN^- attacks the carbon atom of the C–Br bond.
- D The lone pair of electrons on N of CN^- attacks the carbocation formed when the C–Br bond breaks.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

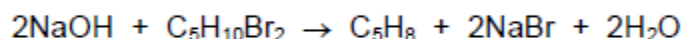


37 In which reactions is the major product formed by a nucleophilic substitution reaction?

- 1 bromoethane + potassium cyanide in ethanol
- 2 bromoethane + ammonia in ethanol under pressure
- 3 bromoethane + hot concentrated sodium hydroxide in ethanol

Q# 945/ AS Chemistry/2021/m/TZ 2/Paper 1/Q# 25//www.SmashingScience.org :o)

25 Dibromopentanes can undergo 'double elimination' reactions to produce hydrocarbons.

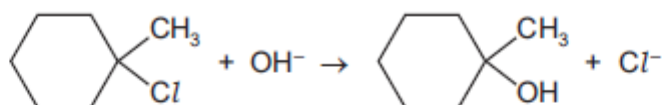


Which isomer produces only one hydrocarbon product?

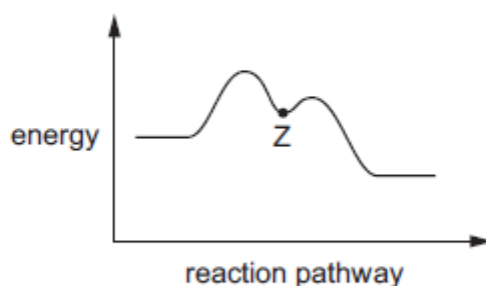
- A 1,5-dibromopentane
- B 1,4-dibromopentane
- C 2,3-dibromopentane
- D 2,4-dibromopentane

Q# 946/ AS Chemistry/2020/w/TZ 1/Paper 1/Q# 25//www.SmashingScience.org :o)

25 1-chloro-1-methylcyclohexane is hydrolysed by heating with NaOH(aq).



The reaction pathway is shown.



One carbon atom in 1-chloro-1-methylcyclohexane is bonded to three other carbon atoms.

What is the charge on this carbon atom at point Z?

- A 1- B δ^- C δ^+ D 1+

Q# 947/ AS Chemistry/2020/w/TZ 1/Paper 1/Q# 23//www.SmashingScience.org :o)

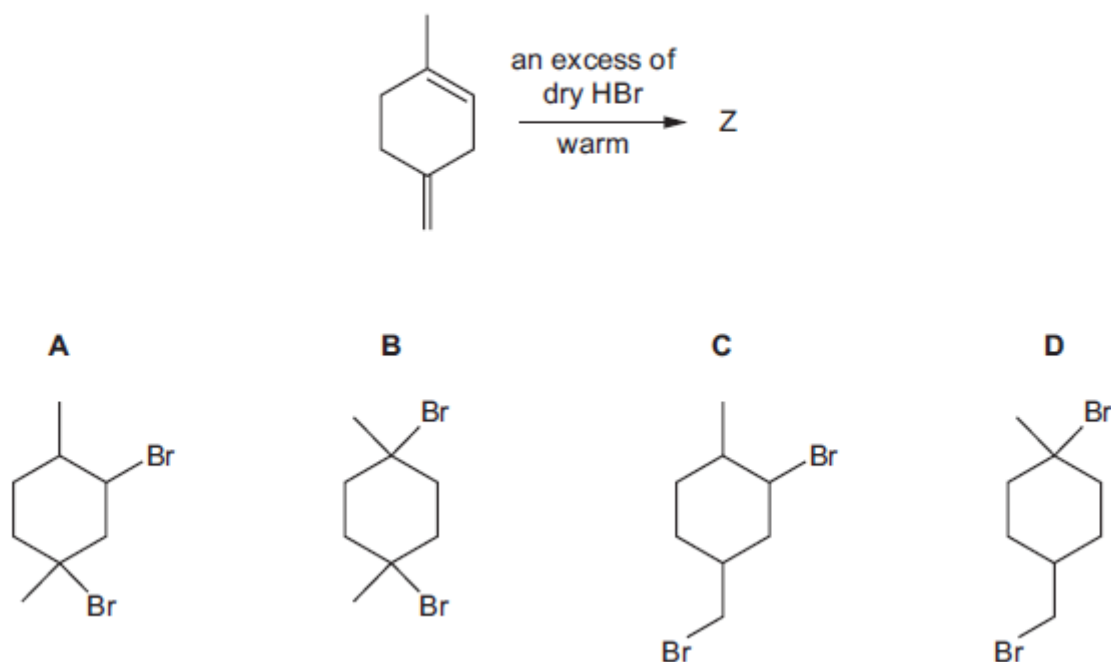
23 Iodoethane, $\text{CH}_3\text{CH}_2\text{I}$, reacts with aqueous silver nitrate at 50°C . A precipitate forms during this reaction.

Which row of the table is correct about this reaction?

	type of organic reaction	colour of precipitate
A	electrophilic substitution	cream
B	electrophilic substitution	yellow
C	nucleophilic substitution	cream
D	nucleophilic substitution	yellow



27 What is the major product Z of the following reaction?



Q# 949/ AS Chemistry/2020/s/TZ 1/Paper 1/Q# 23//www.SmashingScience.org :o)

23 2-bromo-2-methylpentane is a tertiary halogenoalkane.

Which organic products are formed when 2-bromo-2-methylpentane reacts with a hot concentrated ethanolic solution of sodium hydroxide?

- A 2-methylpent-1-ene only
- B 2-methylpent-1-ene and 2-methylpent-2-ene
- C 2-methylpent-2-ene only
- D 2-methylpent-2-ene and 4-methylpent-2-ene

Q# 950/ AS Chemistry/2020/m/TZ 2/Paper 1/Q# 21//www.SmashingScience.org :o)

21 Which row identifies a suitable starting material and reagent that can be used to produce butanenitrile?

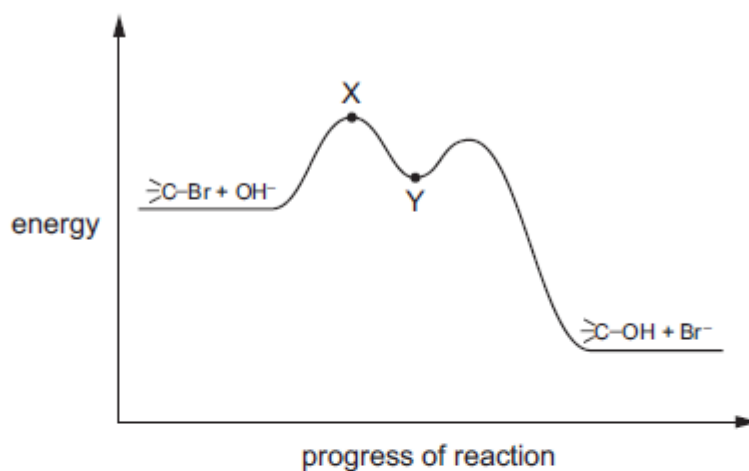
	starting material	reagent
A	$\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$	HCN
B	$\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$	NaCN
C	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$	HCN
D	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$	NaCN

24 Bromoethane and chloroethane are added separately to water. Hydrolysis reactions occur.

Which compound hydrolyses more rapidly and what is the mechanism?

	compound that hydrolyses more rapidly	mechanism
A	bromoethane	electrophilic substitution
B	bromoethane	nucleophilic substitution
C	chloroethane	electrophilic substitution
D	chloroethane	nucleophilic substitution

23 A tertiary bromoalkane, indicated here by >C-Br , reacts with aqueous NaOH. The mechanism has the reaction pathway shown.



Which point in the diagram is correctly identified?

A X is >C^+

B X is $\left[\text{HO} \cdots \text{C} \cdots \text{Br} \right]^-$

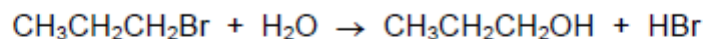
C Y is >C^+

D Y is $\left[\text{HO} \cdots \text{C} \cdots \text{Br} \right]^-$

23 Which reaction is most likely to involve the formation of a positively charged intermediate?

- A 1-bromopentane and warm dilute NaOH(aq)
- B 1-bromo-2,2-dimethylpropane and warm dilute NaOH(aq)
- C 1-bromo-3-methylbutane and warm dilute NaOH(aq)
- D 2-bromo-2-methylbutane and warm dilute NaOH(aq)

24 Bromopropane reacts with water as shown.



Which statement is correct?

- A This is an elimination reaction.
- B This is a hydrolysis reaction.
- C This is a redox reaction.
- D This reaction tends to proceed via the $\text{S}_{\text{N}}1$ mechanism.

Q# 955/ AS Chemistry/2019/m/TZ 2/Paper 1/Q# 23//www.SmashingScience.org :o)

23 Structural isomerism and stereoisomerism should be considered when answering this question.

2-bromopentane is heated with an excess of ethanolic sodium hydroxide.

How many different hydrocarbons are produced?

- A 1
- B 2
- C 3
- D 4

Q# 956/ AS Chemistry/2018/w/TZ 1/Paper 1/Q# 38//www.SmashingScience.org :o)

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

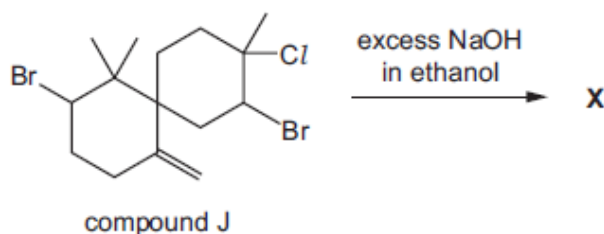
38 The halogenoalkanes listed below all react with NaOH(aq) .

Which reactions proceed mainly by an $\text{S}_{\text{N}}1$ mechanism?

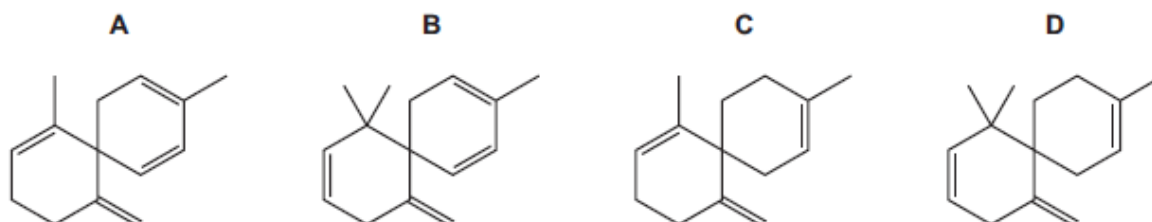
- 1 1-iodopropane
- 2 2-iodo-2-methylpropane
- 3 2-bromo-2-methylbutane

Q# 957/ AS Chemistry/2018/w/TZ 1/Paper 1/Q# 25//www.SmashingScience.org :o)

25 Compound J, $\text{C}_{15}\text{H}_{23}\text{Br}_2\text{Cl}$, is reacted with an excess of a hot concentrated solution of sodium hydroxide in ethanol. One of the products is X.

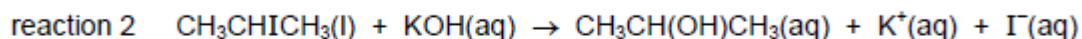
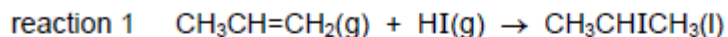


What could be the skeletal formula of X?



Q# 958/ AS Chemistry/2018/w/TZ 1/Paper 1/Q# 23//www.SmashingScience.org :o)

- 23 The conversion of propene to propan-2-ol can be carried out in two stages represented by the equations shown.



How can these two reactions be described?

	reaction 1	reaction 2
A	addition	elimination
B	addition	substitution
C	elimination	substitution
D	substitution	elimination

Q# 959/ AS Chemistry/2018/w/TZ 1/Paper 1/Q# 20//www.SmashingScience.org :o)

- 20 What is true of **every** nucleophile?

- A It attacks a double bond.
- B It donates a lone pair of electrons.
- C It is a single atom.
- D It is negatively charged.

Q# 960/ AS Chemistry/2018/s/TZ 1/Paper 1/Q# 22//www.SmashingScience.org :o)

- 22 Sodium methoxide, $\text{Na}^+\text{CH}_3\text{O}^-$, reacts with 2-chloro-2-methylpropane in a nucleophilic substitution reaction. The nucleophile is the CH_3O^- ion.

Which row is correct?

	intermediate or transition state	product
A	$(\text{CH}_3)_3\text{C}^+$	$(\text{CH}_3)_3\text{COCH}_3$
B	$(\text{CH}_3)_3\text{C}^+$	$(\text{CH}_3)_3\text{CCH}_2\text{OH}$
C	$[\text{HOCH}_2 - - \text{C}(\text{CH}_3)_3 - - \text{Cl}]^-$	$\text{HOCH}_2\text{C}(\text{CH}_3)_3$
D	$[\text{H}_3\text{CO} - - \text{C}(\text{CH}_3)_3 - - \text{Cl}]^-$	$\text{H}_3\text{COC}(\text{CH}_3)_3$

Q# 961/ AS Chemistry/2018/m/TZ 2/Paper 1/Q# 38//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct



38 Organic compound X gives a precipitate when warmed with aqueous silver nitrate. This precipitate dissolves when concentrated aqueous ammonia is added.

What could X be?

- 1 1-bromopropane
- 2 2-chlorobutane
- 3 2-iodo-2-methylpropane

Q# 962/ AS Chemistry/2018/m/TZ 2/Paper 1/Q# 27//www.SmashingScience.org :o)

27 A reaction occurs when a sample of 1-chloropropane is heated under reflux with sodium hydroxide dissolved in ethanol.

Which row is correct?

	type of reaction	name of product
A	elimination	propan-1-ol
B	elimination	propene
C	substitution	propan-1-ol
D	substitution	propene

Q# 963/ AS Chemistry/2018/m/TZ 2/Paper 1/Q# 21//www.SmashingScience.org :o)

21 An organic ion containing a carbon atom with a negative charge is called a carbanion.

An organic ion containing a carbon atom with a positive charge is called a carbocation.

The reaction between aqueous sodium hydroxide and 1-bromobutane proceeds by an S_N2 mechanism.

What is the first step in the mechanism?

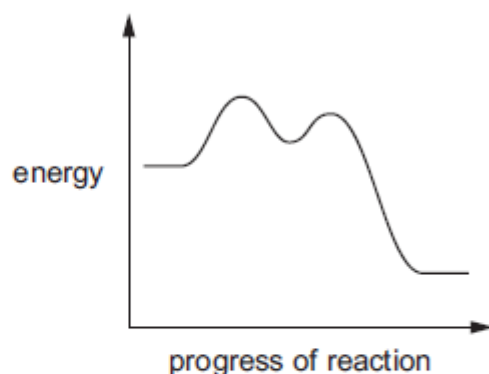
- A attack by a nucleophile on a carbon atom with a partial positive charge
- B heterolytic bond fission followed by attack by an electrophile on a carbanion
- C heterolytic bond fission followed by attack by a nucleophile on a carbocation
- D homolytic bond fission followed by attack by a nucleophile on a carbocation

Q# 964/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 38//

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

38 A reaction pathway diagram is shown.



Which reactions would have this reaction pathway diagram?

- 1 $(\text{CH}_3)_3\text{CBr} + \text{NaOH} \rightarrow (\text{CH}_3)_3\text{COH} + \text{NaBr}$
- 2 $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br} + \text{NaOH} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + \text{NaBr}$
- 3 $(\text{CH}_3)_3\text{CCH}_2\text{CH}_2\text{Cl} + 2\text{NH}_3 \rightarrow (\text{CH}_3)_3\text{CCH}_2\text{CH}_2\text{NH}_2 + \text{NH}_4\text{Cl}$

Q# 965/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 31//

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

31 The definitions of many chemical terms can be illustrated by chemical equations.

Which terms can be illustrated by an equation that includes the formation of a positive ion?

- 1 first ionisation energy
- 2 heterolytic fission of a covalent bond
- 3 enthalpy change of atomisation

Q# 966/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 24//

24 Which organic reaction is an example of nucleophilic substitution?

- A $\text{CH}_3\text{CH}_2\text{Br} + \text{NaOH} \rightarrow \text{CH}_2\text{CH}_2 + \text{H}_2\text{O} + \text{NaBr}$
- B $\text{CH}_3\text{CH}_2\text{Br} + \text{NaOH} \rightarrow \text{CH}_3\text{CH}_2\text{OH} + \text{NaBr}$
- C $\text{CH}_2\text{CH}_2 + \text{HCl} \rightarrow \text{C}_2\text{H}_5\text{Cl}$
- D $\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{HCl}$

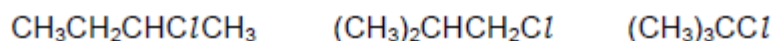
Q# 967/ AS Chemistry/2017/s/TZ 1/Paper 1/Q# 38//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct



- 38 The following statements are about the reaction of NaOH(aq) with the three chloroalkanes shown.



Which statements are correct?

- 1 $(\text{CH}_3)_2\text{CHCH}_2\text{Cl}$ reacts with NaOH(aq) by an $\text{S}_{\text{N}}2$ mechanism.
- 2 The tertiary chloroalkane reacts more quickly than the others because the carbon atom bonded to the Cl atom is more positive in this molecule.
- 3 The Cl atoms in the three chloroalkanes are attacked by OH^- .

Q# 968/ AS Chemistry/2017/m/TZ 2/Paper 1/Q# 38//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

- 38 Which statements help to explain the mechanism of the reaction between 1-chloropropane and ammonia?

- 1 1-chloropropane has a δ^- chlorine atom that forms hydrogen bonds with a δ^+ hydrogen atom in ammonia.
- 2 1-chloropropane is a polar compound with a δ^+ carbon atom.
- 3 There is a lone pair of electrons on the nitrogen atom in ammonia.

Q# 969/ AS Chemistry/2017/m/TZ 2/Paper 1/Q# 22//www.SmashingScience.org :o)

- 22 Which radical is most likely to form by the homolytic fission of one covalent bond in bromochloromethane, CH_2BrCl ?

A $\cdot\text{CH}_2\text{Cl}$ B $\cdot\text{CH}_2\text{Br}$ C $\cdot\text{CHBrCl}$ D $\cdot\text{CH}_2\text{BrCl}$

Q# 970/ AS Chemistry/2016/w/TZ 1/Paper 1/Q# 38//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

- 38 Bromoethane reacts with NaOH in different ways depending on the solvent used.

Which statements about these reactions are correct?

	solvent used	main organic product
1	water	ethane-1,2-diol
2	ethanol	ethene
3	water	ethanol



The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

36 Which types of reaction can occur with 1-bromobutane?

- 1 elimination
- 2 hydrolysis
- 3 free radical substitution

25 A student prepares pentan-1-ol by the alkaline hydrolysis of 1-iodopentane. She gently warms the reaction mixture for 20 minutes.



When the student uses 1-chloropentane to prepare the same alcohol she has to change the condition of the reaction.

Which change in condition should she use and what is the correct reason for its use?

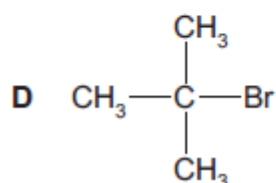
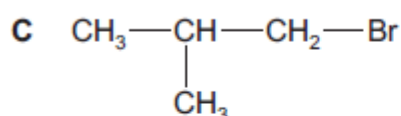
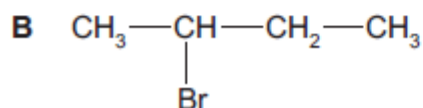
	change in condition	reason
A	heat under reflux	C–Cl bond is more polar than the C–I bond
B	heat under reflux	C–Cl bond is stronger than the C–I bond
C	room temperature	C–Cl bond is more polar than the C–I bond
D	room temperature	C–Cl bond is shorter than the C–I bond



25 Structural isomerism and stereoisomerism should be considered in answering this question.

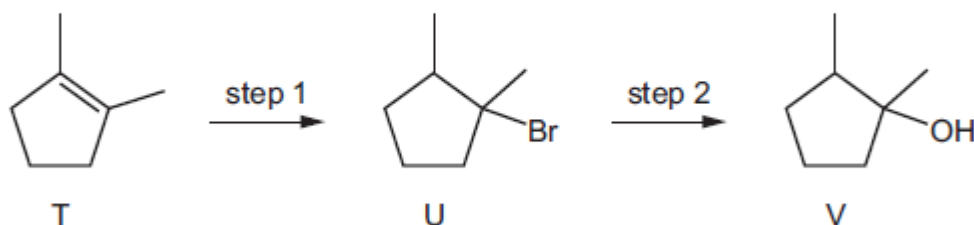
Compound J is reacted with KOH dissolved in ethanol. Three isomeric alkenes with molecular formula C_4H_8 are formed.

What is J?



Q# 974/ AS Chemistry/2016/m/TZ 2/Paper 1/Q# 24//www.SmashingScience.org :o)

24 Hydrogen bromide can be added to T to give compound U. Compound U can be hydrolysed to compound V.



Four students, W, X, Y and Z, made the following statements.

W All the atoms in a molecule of compound T lie in the same plane.

X Compound V contains only one chiral centre.

Y Step 1 is an electrophilic addition reaction.

Z Step 2 is a nucleophilic substitution reaction.

Which two students are correct?

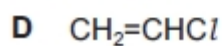
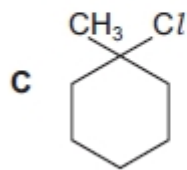
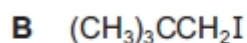
A W and Y

B W and Z

C X and Y

D Y and Z

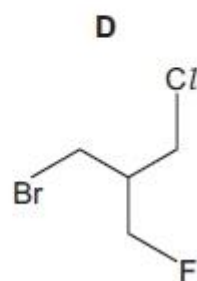
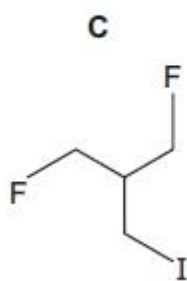
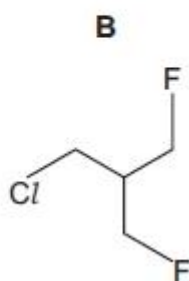
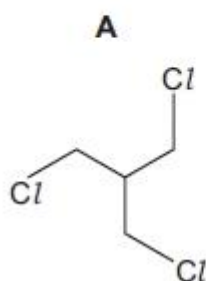
24 Which compound undergoes an S_N1 substitution reaction with NaOH(aq) ?



Q# 976/ AS Chemistry/2015/s/TZ 1/Paper 1/Q# 26//www.SmashingScience.org :o)

26 The presence of a halogen in an organic compound may be detected by warming the organic compound with aqueous silver nitrate.

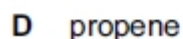
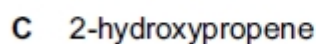
Which compound would be the quickest to produce a precipitate?



Q# 977/ AS Chemistry/2015/s/TZ 1/Paper 1/Q# 25//www.SmashingScience.org :o)

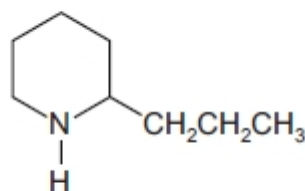
25 2-bromopropane reacts with a hot concentrated solution of sodium hydroxide in ethanol.

Which substance is the major product of this reaction?



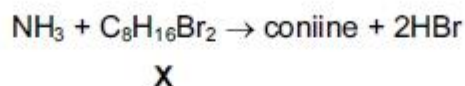
Q# 978/ AS Chemistry/2015/s/TZ 1/Paper 1/Q# 24//www.SmashingScience.org :o)

24 Coniine is the major constituent of the poison 'oil of hemlock'.



coniine

Coniine can be synthesised by reacting ammonia with a dibromo compound, **X**.



What is the name of compound **X**?

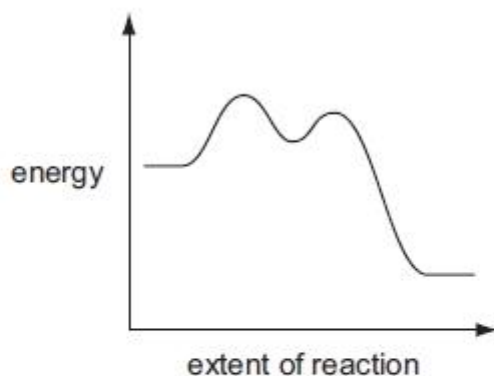
- A** 1,1-dibromo-2-propylcyclopentane
- B** 1,2-dibromo-2-propylcyclopentane
- C** 1,4-dibromooctane
- D** 1,5-dibromooctane

Q# 979/ AS Chemistry/2014/w/TZ 1/Paper 1/Q# 40//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 A reaction pathway diagram is shown.



Which reactions would have this profile?

- 1** $(\text{CH}_3)_3\text{CBr} + \text{NaOH} \rightarrow (\text{CH}_3)_3\text{COH} + \text{NaBr}$
- 2** $\text{CH}_3\text{CH}_2\text{Br} + \text{NaOH} \rightarrow \text{CH}_3\text{CH}_2\text{OH} + \text{NaBr}$
- 3** $(\text{CH}_3)_3\text{CCH}_2\text{CH}_2\text{Cl} + 2\text{NH}_3 \rightarrow (\text{CH}_3)_3\text{CCH}_2\text{CH}_2\text{NH}_2 + \text{NH}_4\text{Cl}$

Q# 980/ AS Chemistry/2014/w/TZ 1/Paper 1/Q# 38//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct



- 38 X is an organic compound that gives a precipitate with aqueous silver nitrate. This precipitate remains undissolved when concentrated aqueous ammonia is added.

What is a possible identity for X?

- 1 iodomethane
- 2 2-bromobutane
- 3 2-chlorobutane

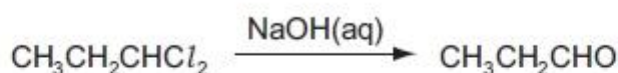
Q# 981/ AS Chemistry/2014/w/TZ 1/Paper 1/Q# 25//www.SmashingScience.org :o)

- 25 In the hydrolysis of bromoethane by aqueous sodium hydroxide, what is the nature of the attacking group and of the leaving group?

	attacking group	leaving group
A	electrophile	electrophile
B	electrophile	nucleophile
C	nucleophile	electrophile
D	nucleophile	nucleophile

Q# 982/ AS Chemistry/2014/w/TZ 1/Paper 1/Q# 22//www.SmashingScience.org :o)

- 22 1,1-dichloropropane reacts with aqueous sodium hydroxide in a series of steps to give propanal.



Which term describes the first step of this reaction?

- A addition
- B elimination
- C oxidation
- D substitution

Q# 983/ AS Chemistry/2014/s/TZ 1/Paper 1/Q# 23//www.SmashingScience.org :o)

- 23 The hydrolysis of 1-chloropropane to produce propan-1-ol is much slower than the corresponding hydrolysis of 1-iodopropane.

Which statement explains this observation?

- A Chlorine is more electronegative than iodine.
- B The bond strength of the C–I bond is less than that of the C–Cl bond.
- C The carbon atom in the C–Cl bond is more δ^+ than that in the C–I bond.
- D The hydrolysis involves a nucleophilic addition reaction.

Q# 984/ AS Chemistry/2013/w/TZ 1/Paper 1/Q# 37//www.SmashingScience.org :o)

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct



- 37 The organic compound **X** gives a precipitate when warmed with aqueous silver nitrate. This precipitate dissolves when concentrated aqueous ammonia is added.

What is a possible identity for **X**?

- 1 1-bromopropane
- 2 2-chlorobutane
- 3 2-iodo,2-methylpropane

Q# 985/ AS Chemistry/2013/w/TZ 1/Paper 1/Q# 24//www.SmashingScience.org :o)

- 24 A carbanion is an organic ion in which a carbon atom has a negative charge. A carbocation is an organic ion in which a carbon atom has a positive charge.

The reaction between aqueous sodium hydroxide and 1-bromobutane proceeds by an S_N2 mechanism.

How should the first step in the mechanism be described?

- A attack by a nucleophile on a carbon atom with a partial positive charge
- B heterolytic bond fission followed by an attack by an electrophile on a carbanion
- C heterolytic bond fission followed by an attack by a nucleophile on a carbocation
- D homolytic bond fission followed by an attack by a nucleophile on a carbocation

Q# 986/ AS Chemistry/2013/s/TZ 1/Paper 1/Q# 24//www.SmashingScience.org :o)

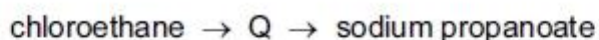
- 24 Aqueous sodium hydroxide reacts with 1-bromopropane to give propan-1-ol.

How should the first step in the mechanism be described?

- A by a curly arrow from a lone pair on the OH^- ion to the $C^{\delta+}$ atom of 1-bromopropane
- B by a curly arrow from the $C^{\delta+}$ atom of 1-bromopropane to the OH^- ion
- C by a curly arrow from the C–Br bond to the C atom
- D by the homolytic fission of the C–Br bond

Q# 987/ AS Chemistry/2013/s/TZ 1/Paper 1/Q# 23//www.SmashingScience.org :o)

- 23 Chloroethane can be used to make sodium propanoate.



The intermediate, Q, is hydrolysed with boiling aqueous sodium hydroxide, to give sodium propanoate.

Which reagent would produce the intermediate, Q, from chloroethane?

- A concentrated ammonia solution
- B dilute sulfuric acid
- C hydrogen cyanide
- D potassium cyanide



The responses **A** to **D** should be selected on the basis of

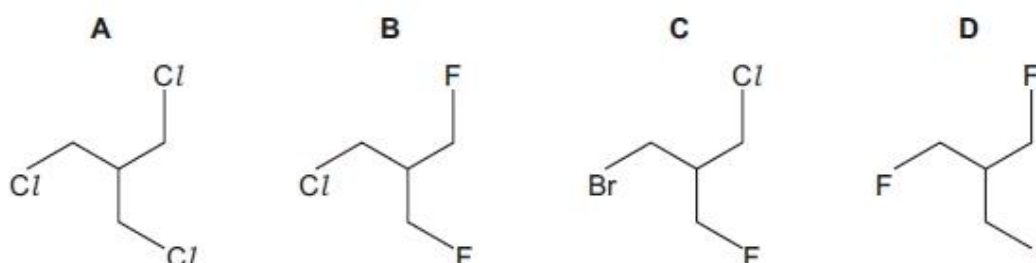
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 How can a good yield of ethylamine be made using bromoethane as starting material?

- 1 by heating bromoethane with an excess of ammonia gas in a sealed tube
- 2 by adding dilute aqueous ammonia to bromoethane at room temperature
- 3 by heating bromoethane under reflux with aqueous ammonium chloride

27 The presence of halogen in an organic compound may be detected by warming the organic compound with aqueous silver nitrate.

Which compound would produce a precipitate quickest?



The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

40 **X** is an organic compound. **X** gives a precipitate with aqueous silver nitrate. Some or all of this precipitate remains undissolved when an excess of dilute aqueous ammonia is added.

What could be the identity of **X**?

- 1 2-chlorobutane
- 2 2-bromobutane
- 3 iodomethane

24 What is involved in the mechanism of the reaction between aqueous sodium hydroxide and 1-bromobutane?

- A attack by a nucleophile on a carbon atom with a partial positive charge
- B heterolytic bond fission and attack by a nucleophile on a carbocation
- C homolytic bond fission and attack by an electrophile on a carbanion
- D homolytic bond fission and attack by a nucleophile on a carbocation



The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

39 Bromoethane undergoes all of the conversions shown.

Which conversions are examples of nucleophilic substitution?

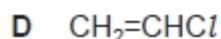
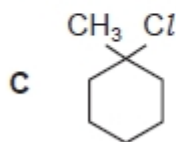
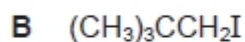
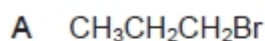
- 1 $\text{C}_2\text{H}_5\text{Br} \rightarrow \text{C}_2\text{H}_5\text{CN}$
- 2 $\text{C}_2\text{H}_5\text{Br} \rightarrow \text{C}_2\text{H}_5\text{OH}$
- 3 $\text{C}_2\text{H}_5\text{Br} \rightarrow \text{C}_2\text{H}_5\text{NH}_2$

22 The formula CH_3 can represent an anion, a cation or a free radical. Species with the molecular formula CH_3 can act as an electrophile, a free radical or a nucleophile depending on the number of outer shell electrons on the central carbon atom.

How many outer shell electrons must be present for CH_3 to act in these different ways?

	CH_3 as an electrophile	CH_3 as a free radical	CH_3 as a nucleophile
A	6	7	8
B	6	8	7
C	7	6	8
D	8	7	6

28 Which compound undergoes an $\text{S}_{\text{N}}1$ substitution reaction?



The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct



39 Which reactions are examples of nucleophilic substitution?

- 1 $\text{CH}_3\text{CH}_2\text{Br} + \text{OH}^- \rightarrow \text{CH}_3\text{CH}_2\text{OH} + \text{Br}^-$
- 2 $\text{CH}_3\text{I} + \text{H}_2\text{O} \xrightarrow{\text{H}^+} \text{CH}_3\text{OH} + \text{HI}$
- 3 $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{NH}_3 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2 + \text{HCl}$

Q# 996/ AS Chemistry/2010/s/TZ 1/Paper 1/Q# 28//www.SmashingScience.org :o)

28 When an isomer Y of molecular formula $\text{C}_4\text{H}_9\text{Br}$ undergoes hydrolysis in aqueous alkali to form an alcohol $\text{C}_4\text{H}_9\text{OH}$, the rate of reaction is found to be unaffected by changes in the concentration of OH^- ions present.

Which is the most likely molecular structure of Y?

- A $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$
- B $\text{CH}_3\text{CH}_2\text{CHBrCH}_3$
- C $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$
- D $(\text{CH}_3)_3\text{CBr}$

Q# 997/ AS Chemistry/2009/w/TZ 1/Paper 1/Q# 37//www.SmashingScience.org :o)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

37 Which statements are true for an $\text{S}_{\text{N}}2$ reaction?

- 1 One bond is broken as another bond is formed.
- 2 The formation of a transition state involves the collision of two molecules or ions.
- 3 A carbon atom in the transition state is bonded, either fully or partially, to five other atoms.

Q# 998/ AS Chemistry/2009/s/TZ 1/Paper 1/Q# 24//www.SmashingScience.org :o)

24 In the hydrolysis of bromoethane by aqueous sodium hydroxide, what is the nature of the attacking group and of the leaving group?

	attacking group	leaving group
A	electrophile	electrophile
B	electrophile	nucleophile
C	nucleophile	electrophile
D	nucleophile	nucleophile

Q# 999/ AS Chemistry/2009/s/TZ 1/Paper 1/Q# 21//www.SmashingScience.org :o)

21 What always applies to a nucleophile?

- A It attacks a double bond.
- B It has a lone pair of electrons.
- C It is a single atom.
- D It is negatively charged.



Q# 935/ AS Chemistry/2022/w/TZ 1/Paper 1/Q#

37//www.SmashingScience.org :o)

37 | D

Q# 936/ AS Chemistry/2022/s/TZ 1/Paper 1/Q#

32//www.SmashingScience.org :o)

32 | C

Q# 937/ AS Chemistry/2022/s/TZ 1/Paper 1/Q#

31//www.SmashingScience.org :o)

31 | C

Q# 938/ AS Chemistry/2022/m/TZ 2/Paper 1/Q#

30//www.SmashingScience.org :o)

30 | A

Q# 939/ AS Chemistry/2022/m/TZ 2/Paper 1/Q#

29//www.SmashingScience.org :o)

29 | B

Q# 940/ AS Chemistry/2021/w/TZ 1/Paper 1/Q#

23//www.SmashingScience.org :o)

23 | B

Q# 941/ AS Chemistry/2021/w/TZ 1/Paper 1/Q#

22//www.SmashingScience.org :o)

22 | D

Q# 942/ AS Chemistry/2021/s/TZ 1/Paper 1/Q#

37//www.SmashingScience.org :o)

37 | A

Q# 943/ AS Chemistry/2021/s/TZ 1/Paper 1/Q#

20//www.SmashingScience.org :o)

20 | A

Q# 944/ AS Chemistry/2021/m/TZ 2/Paper 1/Q#

37//www.SmashingScience.org :o)

37 | B

Q# 945/ AS Chemistry/2021/m/TZ 2/Paper 1/Q#

25//www.SmashingScience.org :o)

25 | A

Q# 946/ AS Chemistry/2020/w/TZ 1/Paper 1/Q#

25//www.SmashingScience.org :o)

25 | D

Q# 947/ AS Chemistry/2020/w/TZ 1/Paper 1/Q#

23//www.SmashingScience.org :o)

23 | D

Q# 948/ AS Chemistry/2020/s/TZ 1/Paper 1/Q#

27//www.SmashingScience.org :o)

27 | B

Q# 949/ AS Chemistry/2020/s/TZ 1/Paper 1/Q#

23//www.SmashingScience.org :o)

23 | B

Q# 950/ AS Chemistry/2020/m/TZ 2/Paper 1/Q#

21//www.SmashingScience.org :o)

21 | B

Q# 951/ AS Chemistry/2019/w/TZ 1/Paper 1/Q#

24//www.SmashingScience.org :o)

24 | B

Q# 952/ AS Chemistry/2019/w/TZ 1/Paper 1/Q#

23//www.SmashingScience.org :o)

23 | C

Q# 953/ AS Chemistry/2019/s/TZ 1/Paper 1/Q#

23//www.SmashingScience.org :o)

23 | D

Q# 954/ AS Chemistry/2019/m/TZ 2/Paper 1/Q#

24//www.SmashingScience.org :o)

24 | B

Q# 955/ AS Chemistry/2019/m/TZ 2/Paper 1/Q#

23//www.SmashingScience.org :o)

23 | C

Q# 956/ AS Chemistry/2018/w/TZ 1/Paper 1/Q#

38//www.SmashingScience.org :o)

38 | C

Q# 957/ AS Chemistry/2018/w/TZ 1/Paper 1/Q#

25//www.SmashingScience.org :o)

25 | B

Q# 958/ AS Chemistry/2018/w/TZ 1/Paper 1/Q#

23//www.SmashingScience.org :o)

23 | B

Q# 959/ AS Chemistry/2018/w/TZ 1/Paper 1/Q#

20//www.SmashingScience.org :o)

20 | B

Q# 960/ AS Chemistry/2018/s/TZ 1/Paper 1/Q#

22//www.SmashingScience.org :o)

22 | A

Q# 961/ AS Chemistry/2018/m/TZ 2/Paper 1/Q#

38//www.SmashingScience.org :o)

38 | B

Q# 962/ AS Chemistry/2018/m/TZ 2/Paper 1/Q#

27//www.SmashingScience.org :o)

27 | B

Q# 963/ AS Chemistry/2018/m/TZ 2/Paper 1/Q#

21//www.SmashingScience.org :o)

21 | A

Q# 964/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 38//

38 | D

Q# 965/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 31//

31 | B

Q# 966/ AS Chemistry/2017/w/TZ 1/Paper 1/Q# 24//

24 | B



Q# 967/ AS Chemistry/2017/s/TZ 1/Paper 1/Q# 38//www.SmashingScience.org :o)

38 | D

Q# 968/ AS Chemistry/2017/m/TZ 2/Paper 1/Q# 38//www.SmashingScience.org :o)

38 | C

Q# 969/ AS Chemistry/2017/m/TZ 2/Paper 1/Q# 22//www.SmashingScience.org :o)

22 | A

Q# 970/ AS Chemistry/2016/w/TZ 1/Paper 1/Q# 38//www.SmashingScience.org :o)

38 C

Q# 971/ AS Chemistry/2016/s/TZ 1/Paper 1/Q# 36//www.SmashingScience.org :o)

36 A

Q# 972/ AS Chemistry/2016/s/TZ 1/Paper 1/Q# 25//www.SmashingScience.org :o)

25 B

Q# 973/ AS Chemistry/2016/m/TZ 2/Paper 1/Q# 25//www.SmashingScience.org :o)

25 B

Q# 974/ AS Chemistry/2016/m/TZ 2/Paper 1/Q# 24//www.SmashingScience.org :o)

24 D

Q# 975/ AS Chemistry/2015/w/TZ 1/Paper 1/Q# 24//www.SmashingScience.org :o)

24 C

Q# 976/ AS Chemistry/2015/s/TZ 1/Paper 1/Q# 26//www.SmashingScience.org :o)

26 C

Q# 977/ AS Chemistry/2015/s/TZ 1/Paper 1/Q# 25//www.SmashingScience.org :o)

25 D

Q# 978/ AS Chemistry/2015/s/TZ 1/Paper 1/Q# 24//www.SmashingScience.org :o)

24 D

Q# 979/ AS Chemistry/2014/w/TZ 1/Paper 1/Q# 40//www.SmashingScience.org :o)

40 D

Q# 980/ AS Chemistry/2014/w/TZ 1/Paper 1/Q# 38//www.SmashingScience.org :o)

38 D

Q# 981/ AS Chemistry/2014/w/TZ 1/Paper 1/Q# 25//www.SmashingScience.org :o)

25 D

Q# 982/ AS Chemistry/2014/w/TZ 1/Paper 1/Q# 22//www.SmashingScience.org :o)

21 D

Q# 983/ AS Chemistry/2014/s/TZ 1/Paper 1/Q# 23//www.SmashingScience.org :o)

23 B

Q# 984/ AS Chemistry/2013/w/TZ 1/Paper 1/Q# 37//www.SmashingScience.org :o)

37 B

Q# 985/ AS Chemistry/2013/w/TZ 1/Paper 1/Q# 24//www.SmashingScience.org :o)

24 A

Q# 986/ AS Chemistry/2013/s/TZ 1/Paper 1/Q# 24//www.SmashingScience.org :o)

24 A

Q# 987/ AS Chemistry/2013/s/TZ 1/Paper 1/Q# 23//www.SmashingScience.org :o)

23 D

Q# 988/ AS Chemistry/2012/s/TZ 1/Paper 1/Q# 40//www.SmashingScience.org :o)

40 D

Q# 989/ AS Chemistry/2012/s/TZ 1/Paper 1/Q# 27//www.SmashingScience.org :o)

27 D

Q# 990/ AS Chemistry/2011/w/TZ 1/Paper 1/Q# 40//www.SmashingScience.org :o)

40 C

Q# 991/ AS Chemistry/2011/w/TZ 1/Paper 1/Q# 24//www.SmashingScience.org :o)

24 A

Q# 992/ AS Chemistry/2011/s/TZ 1/Paper 1/Q# 39//www.SmashingScience.org :o)

39 A

Q# 993/ AS Chemistry/2011/s/TZ 1/Paper 1/Q# 22//www.SmashingScience.org :o)

22 A

Q# 994/ AS Chemistry/2010/w/TZ 1/Paper 1/Q# 28//www.SmashingScience.org :o)

28 C

Q# 995/ AS Chemistry/2010/s/TZ 1/Paper 1/Q# 39//www.SmashingScience.org :o)

39 A

Q# 996/ AS Chemistry/2010/s/TZ 1/Paper 1/Q# 28//www.SmashingScience.org :o)

28 D

Q# 997/ AS Chemistry/2009/w/TZ 1/Paper 1/Q# 37//www.SmashingScience.org :o)

37 A

Q# 998/ AS Chemistry/2009/s/TZ 1/Paper 1/Q# 24//www.SmashingScience.org :o)

24 D

Q# 999/ AS Chemistry/2009/s/TZ 1/Paper 1/Q# 21//www.SmashingScience.org :o)

21 B

